

About the Covers

Front cover: The determination of molecular structures by cryo-electron microscopy is a major new advance, highlighted in this Seventh Edition, and it has been chosen as the basis for our cover design. One of the main benefits of the technique has been the ability to determine the structure of integral membrane protein complexes that are either hard or impossible to crystallize. Richard Henderson at the MRC Laboratory of Molecular Biology in Cambridge, England (who shared the Nobel Prize in Chemistry in 2017 for cryoEM), suggested six recent atomic structures, all determined by cryoEM, all integral membrane proteins, and all resolved to better than 0.4 nm. Simple images of each of these molecules, which contain multiple subunits, have been drawn to scale and inserted as graphic elements in their “silver” membranes. The six molecular structures (together with their corresponding PDB codes) are as follows, starting clockwise from top left:

1. The voltage-gated sodium channel (see Figure 11–30) from the desert bush spider, a target for many neurotoxins including puffer fish toxin. (PDB code: 6A95.)
2. The human calcitonin-gene-related peptide (CGRP) receptor in complex with its G-protein-coupled receptor (GPCR). (PDB code: 6E3Y.)
3. The light-harvesting reaction center complex (LH1-RC) from the purple bacterium *Blastochloris viridis*. Unusually, it absorbs infrared light. In addition to the reaction center and multiple cofactors, the light-harvesting ring contains 17 alpha, 17 beta, and 16 gamma polypeptides. (PDB code: 6ET5.)
4. The six connexins in the hemichannel of a human connexin assembly; see Figure 19–25. (PDB code: 6L3T.)
5. The fungal class D GPCR from *Saccharomyces cerevisiae* in complex with the dimeric pheromone receptor, Ste2. (PDB code: 7AD3.)
6. The ubiquinol oxidase complex from *Escherichia coli*. (PDB code: 6WTI.)

Back cover: Instead of the more conventional pictures of the book’s authors that appeared on the back covers in the past, we have chosen for the Seventh Edition to play with the idea of a “genetic portrait of each author.” Issues around the public disclosure of personal health information mean that we cannot use our own individual DNA sequences, so instead we have chosen a short 1000-base-pair region of the human genome that contains common, naturally occurring, combinatorial variants and have presented them as if each might have come from one of the authors. The sequence chosen comes from an enhancer for a gene that codes for one of the subunits of the exocyst complex. This complex mediates the tethering of secretory vesicles to the plasma membrane prior to fusion (see Figure 13–21). Each of the four nucleotides in the DNA sequence has been color coded as shown in Figure 4–5, and each author’s 1-kb sequence (100 lines of 10 bases per line) can be read from top left. The seven variants shown include a five-nucleotide deletion (white box) and several single-nucleotide exchanges nearer the end of the sequence (outlined in black). We thank Sai Zhang and Michael Snyder for kindly providing the seven sequences.

The genetic code					
1st position (5' end)	2nd position				3rd position (3' end)
↓	U	C	A	G	↓
U	Phe	Ser	Tyr	Cys	U
	Phe	Ser	Tyr	Cys	C
	Leu	Ser	STOP	STOP	A
	Leu	Ser	STOP	Trp	G
C	Leu	Pro	His	Arg	U
	Leu	Pro	His	Arg	C
	Leu	Pro	Gln	Arg	A
	Leu	Pro	Gln	Arg	G
A	Ile	Thr	Asn	Ser	U
	Ile	Thr	Asn	Ser	C
	Ile	Thr	Lys	Arg	A
	Met	Thr	Lys	Arg	G
G	Val	Ala	Asp	Gly	U
	Val	Ala	Asp	Gly	C
	Val	Ala	Glu	Gly	A
	Val	Ala	Glu	Gly	G

Amino acids and their symbols			Codons
A	Ala	Alanine	GCA GCC GCG GCU
C	Cys	Cysteine	UGC UGU
D	Asp	Aspartic acid	GAC GAU
E	Glu	Glutamic acid	GAA GAG
F	Phe	Phenylalanine	UUC UUU
G	Gly	Glycine	GGA GGC GGG GGU
H	His	Histidine	CAC CAU
I	Ile	Isoleucine	AUA AUC AUU
K	Lys	Lysine	AAA AAG
L	Leu	Leucine	UUA UUG CUA CUC CUG CUU
M	Met	Methionine	AUG
N	Asn	Asparagine	AAC AAU
P	Pro	Proline	CCA CCC CCG CCU
Q	Gln	Glutamine	CAA CAG
R	Arg	Arginine	AGA AGG CGA CGC CGG CGU
S	Ser	Serine	AGC AGU UCA UCC UCG UCU
T	Thr	Threonine	ACA ACC ACG ACU
V	Val	Valine	GUA GUC GUG GUU
W	Trp	Tryptophan	UGG
Y	Tyr	Tyrosine	UAC UAU